

# Technical Manual: Assembly, Firmware, and eDNA Protocol BioOlympia/GlyxonBiolabs

## Bio-Sniffer V3.0: Autonomous Environmental DNA Capture & Multivariable Instrumentation

### Technical Manual, CAD Assembly & eDNA Protocol

## Open-Source Environmental Monitoring Network for Permafrost eDNA Capture & Pathogen Surveillance

- **Microcontroller:** Adafruit Feather ESP32-S3 TFT
- **Firmware Stack:** CircuitPython Core (FSM)
- **Release Date:** August 2026

### 1. Scientific Context & Threat Matrix (Permafrost eDNA)

Accelerated climate change in polar and subpolar latitudes causes rapid thawing of **permafrost**. This exposes ancient organic layers containing dormant microorganisms, viral particles, and preserved environmental DNA (**eDNA**). The **Bio-Sniffer V4.0** is engineered as a decentralized, low-cost monitoring node designed to capture airborne bioaerosols and eDNA fragments originating from thawing polar soils before community exposure occurs.

### Key Case Studies in Literature

- **Siberian Anthrax Outbreak (Yamal, 2016):** Heatwaves thawed an infected reindeer carcass frozen since 1941, releasing viable *Bacillus anthracis* spores that caused widespread reindeer mortality and human hospitalizations.
- **Giant Viruses Revival (2014–2015):** Isolation of viable *Pithovirus sibericum* and *Mollivirus sibericum* from over 30,000-year-old Siberian permafrost layers.
- **Pandoravirus yedoma (2023):** Recovery of infectious viral particles and intact genomic material from permafrost cores dating back over 48,500 years.

# 1. Executive Summary and Venturi Hydrodynamic Architecture

The Bio-Sniffer eDNA is a decentralized, open-hardware device designed for active suction, physical capture of airborne environmental DNA, and multivariable microclimatic telemetry. The prototype's core integrates an Adafruit Feather ESP32-S3 TFT microcontroller mounted on a perforated PCB development board, featuring I2C sensors, ambient luminance reading, and dual LiPo autonomous power management.

Its mechanical architecture employs a hydrodynamic "Venturi effect" to create a constant negative pressure zone in the suction throat via a 3010mm 5V DC axial blower. Ambient air flows from the intake nozzle, passing through a multi-stage filtration system combining coarse mechanical retention and a 25mm (0.22  $\mu$ m) membrane filter, and is channeled toward a desiccant chamber with silica gel to prevent enzymatic/hydrolytic degradation of the captured nucleic acids. Physical separation isolates the thermal sensor modules from the heat emitted by the CPU and the Feather's IPS TFT screen.

## 2. Complete Bill of Materials (BOM)

### 2.1 Electronic Components and Power

| Component | Specification / I2C Address / Pin                      | Qty. | System Function                            |
|-----------|--------------------------------------------------------|------|--------------------------------------------|
| Main MCU  | Adafruit Feather ESP32-S3 TFT (Dual Core, IPS Display) | 1    | Processing, FSM state machine, TFT screen. |

|                               |                                                   |   |                                                                             |
|-------------------------------|---------------------------------------------------|---|-----------------------------------------------------------------------------|
| <b>LiPo Batteries</b>         | Qimoo 606090 3.7V 4000mAh (JST PH2.0mm)           | 2 | Parallel power supply for extended autonomy.                                |
| <b>Terminal Blocks</b>        | Dual screw terminals salvaged from MOSFET module  | 2 | Interconnection and mechanical distribution of the power bus.               |
| <b>Boost Converter Module</b> | MT3608 DC-DC Step-Up (Pre-calibrated to 5.0V)     | 1 | Conversion from 3.7V LiPo to 5.0V for the axial blower.                     |
| <b>MOSFET Driver</b>          | N-Channel MOSFET (2N7000/IRFZ44N) + 10k Pull-down | 1 | Fan switching from GPIO26/D5.                                               |
| <b>Air Quality Sensor</b>     | ENS160 + AHT21 (I2C: 0x53/0x38)                   | 1 | Measurement of eCO <sub>2</sub> , TVOC, Temperature, and Relative Humidity. |
| <b>Barometric Sensor</b>      | BMP280 Barometric (I2C: 0x76)                     | 1 | Measurement of atmospheric pressure and altitude.                           |
| <b>LiPo Fuel Gauge</b>        | MAX17048 Fuel Gauge (I2C: 0x36)                   | 1 | Direct monitoring of battery percentage and voltage.                        |
| <b>Luminance Sensor</b>       | GL5516 Photoresistor + 10k Resistor (Pin A0)      | 1 | Voltage divider for ambient light reading.                                  |
| <b>Axial Fan</b>              | 3010mm DC 5V (Connected to MT3608 VOUT+)          | 1 | Generation of active Venturi suction flow.                                  |

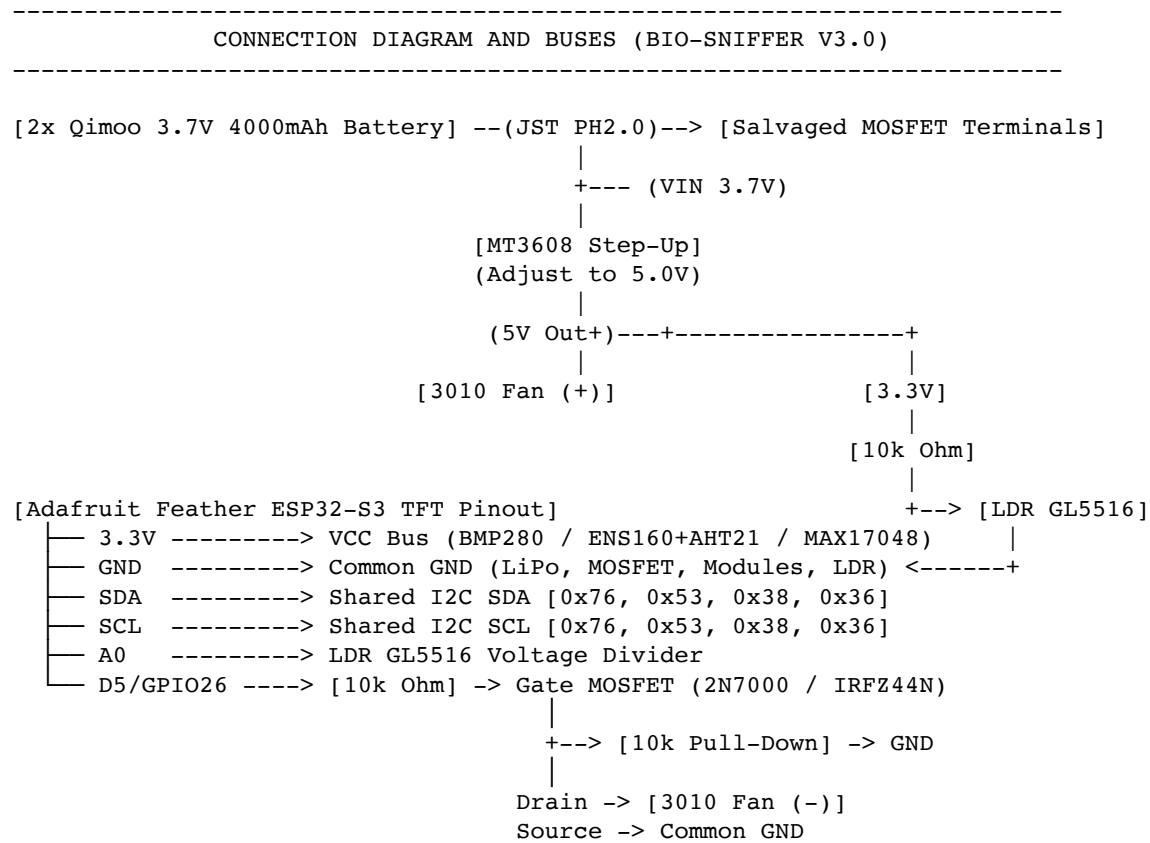
## 2.2 Mechanical Parts and 3D Prints

| 3D File           | Format       | Material     | Structural Description and Function                       |
|-------------------|--------------|--------------|-----------------------------------------------------------|
| venturibody.stl   | STL/<br>STEP | PETG/<br>PLA | Acceleration throat and main body for the Venturi effect. |
| Fan chamber 1.stl | STL/<br>STEP | PETG/<br>PLA | Internal chamber for the 3010mm blower and assembly rack. |

|                              |              |              |                                                              |
|------------------------------|--------------|--------------|--------------------------------------------------------------|
| 25mm filter holder.stl       | STL/<br>STEP | PETG/<br>TPU | Retaining support for the fine membrane disk (0.22 $\mu$ m). |
| main_outer_enclosure.stl     | STL/<br>STEP | PETG/<br>ABS | Translucent outer chassis with ventilation grilles.          |
| end exhaust.stl              | STL/<br>STEP | PETG/<br>PLA | Rear exhaust and air flow expulsion nozzle.                  |
| AMS Dryer pod indent lid.stl | STL/<br>STEP | PETG/<br>PLA | Sealed top cover/lid for the desiccant bay.                  |
| filter holder2-silica.stl    | STL/<br>STEP | PETG/<br>TPU | Intermediate grille and container for silica gel beads.      |

**CRITICAL POWER CALIBRATION WARNING** Before soldering or connecting the VOUT+/VOUT- output of the MT3608 step-up module to the 3010mm fan or the Feather ESP32-S3, power the module with the 3.7V battery and turn the fine potentiometer with a multimeter until it measures **exactly 5.0V DC**. Uncalibrated voltage will destroy the blower or overheat the electronics.

### 3. Wiring Diagram and Electrical Schematic



### 4. CircuitPython Firmware Architecture (Finite State Machine)

The control code is structured as a non-blocking Finite State Machine (FSM) running on CircuitPython Core for the Feather ESP32-S3 TFT, divided into 3 primary states:

- **STATE\_IDLE (0):** Passive monitoring loop. Samples the I2C sensors (BMP280, ENS160, AHT21, LDR, MAX17048) every 5 seconds. Evaluates the biological alert window:  $\text{Temperature} \leq 30.0^{\circ}\text{C}$  **AND**  $\text{Relative Humidity} \geq 55.0\%$ . If met, switches to STATE\_SAMPLING.

- **STATE\_SAMPLING (1):** Activates the GPIO pin (D5) HIGH (3.3V), exciting the MOSFET gate and starting fan suction at 5V. Maintains active capture for a strict interval of 20 seconds (**SAMPLING\_DURATION**). Features a hysteresis abort rule ( $\text{Temp} > 31.5^{\circ}\text{C}$  or  $\text{Hum} < 52.0\%$ ). Upon completion, turns off the fan and transitions to **STATE\_COOLDOWN**.
- **STATE\_COOLDOWN (2):** Hardware lockout period of 10 seconds (**COOLDOWN\_DURATION**) to prevent rapid fluctuations, overheating, and allow sensor stabilization before returning to **STATE\_IDLE**.

## 4.1 Main Source Code (**code\_climate\_fsm.py**)

(Python)

```
import time
import board
import digitalio
from analogio import AnalogIn
import adafruit_bmp280
import adafruit_ahtx0
import adafruit_ens160
import adafruit_max1704x

# Pinout & Hardware
FAN_PIN = board.D5
PHOTORESISTOR_PIN = board.A0
i2c = board.STEMMA_I2C()
pressure_sensor = adafruit_bmp280.Adafruit_BMP280_I2C(i2c, 0x76)
temp_hum_sensor = adafruit_ahtx0.AHTx0(i2c)
gas_sensor = adafruit_ens160.ENS160(i2c)
battery_sensor = adafruit_max1704x.MAX17048(i2c)

fan = digitalio.DigitalInOut(FAN_PIN)
fan.direction = digitalio.Direction.OUTPUT
fan.value = False
light_sensor = AnalogIn(PHOTORESISTOR_PIN)
```

```

# Thresholds & Timing
TEMP_MAX_THRESHOLD = 30.0
HUM_MIN_THRESHOLD = 55.0
HYSTERESIS_TEMP = 1.5
HYSTERESIS_HUM = 3.0
POLL_INTERVAL = 5.0
COOLDOWN_DURATION = 10.0
SAMPLING_DURATION = 20.0

STATE_IDLE, STATE_SAMPLING, STATE_COOLDOWN = 0, 1, 2
current_state = STATE_IDLE
prev_t = time.monotonic()
cooldown_start = 0.0
sampling_start = 0.0

while True:
    now = time.monotonic()
    if now - prev_t > POLL_INTERVAL:
        prev_t = now
        try:
            current_temp = temp_hum_sensor.temperature
            current_hum = temp_hum_sensor.relative_humidity
            current_pressure = pressure_sensor.pressure
            gas_sensor.temperature_compensation = current_temp
            gas_sensor.humidity_compensation = current_hum
            current_gas_eco2 = gas_sensor.eCO2
            current_ldr = (light_sensor.value / 65536) * 100
            current_batt_percent = battery_sensor.cell_percent
            current_batt_voltage = battery_sensor.cell_voltage
        except Exception as e:
            print("Sensor Error:", e)
            continue

        print("Temp: %.1f C | Hum: %.1f %% | eCO2: %.1f ppm | Light: %.2f %%"
              %
              (current_temp, current_hum, current_gas_eco2, current_ldr))

    if current_state == STATE_IDLE:
        if current_temp < TEMP_MAX_THRESHOLD and current_hum >=
HUM_MIN_THRESHOLD:
            print("-> [ALERT] Target window hit. Commencing capture...")
            fan.value = True
            sampling_start = time.monotonic()
            current_state = STATE_SAMPLING
    elif current_state == STATE_SAMPLING:
        if (current_temp > (TEMP_MAX_THRESHOLD + HYSTERESIS_TEMP) or
            current_hum < (HUM_MIN_THRESHOLD - HYSTERESIS_HUM)):
            print("-> [ABORT] Conditions degraded.")
            fan.value = False
            cooldown_start = time.monotonic()
            current_state = STATE_COOLDOWN
    elif time.monotonic() - sampling_start > SAMPLING_DURATION:
        print("-> Sampling complete. Cooldown mode.")
        fan.value = False

```

```

        cooldown_start = time.monotonic()
        current_state = STATE_COOLDOWN
    elif current_state == STATE_COOLDOWN:
        if time.monotonic() - cooldown_start > COOLDOWN_DURATION:
            print("-> Cooldown done. Re-entering IDLE.")
            current_state = STATE_IDLE

```

## 5. Recovery, Genomic Lysis, and 16S rRNA Analysis Workflow

- **STEP 1: Filter and Mesh Recovery**

Upon completing the active suction cycle, remove the `AMS Dryer pod indent lid.stl` cover and the silica tray. Using sterilized tweezers, extract the 25mm fine nylon/nitrocellulose membrane disk from `25mm filter holder.stl` (Fine Filter) and the coarse pre-filter mesh. Place the fine filter in Tube A and the pre-filter in Tube B with Pre-Lysis Buffer.

- **STEP 2: Protocolized Total DNA Extraction**

Add Lysis Buffer to the samples. Vortex vigorously to detach the eDNA biomass adhered to the fibers. Incubate at 65°C for enzymatic digestion and cell lysis. Centrifuge at high speed to precipitate solid debris ("pellet debris"). Pass the purified supernatant through a silica affinity column, apply Wash Buffer, and elute the concentrated eDNA.

- **STEP 3: 16S rRNA Molecular Marker and Bioinformatic Identification**

For taxonomic characterization of captured bacterial and archaeal communities, the gene encoding the 30S ribosomal subunit is amplified (16S rRNA, ~1500 bp):

- **Conserved Regions:** Universal primer binding sites ensuring multi-specific amplification.
  - **Hypervariable Regions (V1-V9):** Variable genomic segments acting as a "fingerprint" to differentiate genera and species with high resolution.

- **Bioinformatic Alignment:** Sequencing reads are compared using bioinformatic pipelines against curated databases (SILVA, GTDB, EzBioCloud, NCBI). This flow enables the identification of extremophilic/polar species or infiltrated permafrost bacteria (e.g., *Psychrobacter cryohalolentis*, *Methanosarcina permafrosti*).

## **FIELD VERIFICATION AND OPERATION CHECKLIST**

1. Confirm calibration of the MT3608 boost converter to exactly 5.0V.
2. Check I2C response on the TFT screen (BMP280, ENS160, AHT21, MAX17048).
3. Verify presence of dry silica gel in `filter_holder2-silica.stl`.
4. Confirm proper seating of the 25mm filter without perimeter air leaks.